COURSE 2 · SECTION 3 · EXERCISE

Where does the NPD schedule slip?

Section 3 · New product development: concept to release

Start: Section04_Start.pbix or your Engxtech_S03.pbix; save a copy as Engxtech_S03_Exercise.pbix

Time: about 20 min

Data: the course data folder

DESIGN QUESTION

Across all 16 part revisions, which NPD stage misses its planned end most often, where are the slip days, and how much of the P300-B drawing delay came from Checking?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

NPD_Stages has 12 stages for each part revision (192 rows), with planned and actual start/end, the owner and the gate result. Stages 1 to 10 end at Release; Prototype & validation and Production handover start at release. The video added the columns Plan Days, Actual Days, Slip Days and On Time, and the measures On-time Gate %, Median Plan Days, Median Actual Days and Total Slip Days.

Your tasks

1. In your exercise copy, check the video's columns on NPD_Stages: Plan Days, Actual Days, Slip Days (= max of 0 and Actual Days - Plan Days) and On Time (1 if ActualEnd is on or before PlannedEnd). Stage is sorted by StageNo.
2. Right-click NPD_Stages > New measure **Rework Gates**: the number of rows with GateResult = Rework.
3. Add a table by Stage with On-time Gate %, Total Slip Days, Median Plan Days and Median Actual Days. Which stage has the lowest on-time %, and which the most slip days?
4. On the NPD page, pick P300-B in the PartRev slicer and read the Gantt bars (Actual Offset with no fill + Actual Days). Read the P300-B slip by stage and add up stages 1 to 10.
5. Count the Rework gates.

Check your answer

Check	Expected
On-time gates, all stages	**56.3 % (108/192)**
On-time gates, Checking	**37.5 % (6/16)**
Stage with the most slip days	**Checking 31 d**
Slip days, all stages	**110 d**
Checking median planned / actual days	**4.0 / 6.0**
P300-B slip in Checking	**10 d**
P300-B slip, stages 1-10 (= drawing lateness)	**19 d**
Rework gates	**16**

Hint. Slip Days counts the stage's own overrun only. A stage that starts late because an earlier stage slipped shows 0, so the stage sum for P300-B up to Release equals the drawing lateness.

Stuck? The worked solution with the full DAX and M is in Solutions/S03_Solution.pdf.